

		acceleration = 9.77 m s^{-2}	B1	[2]
2	(a)	(i) a, ω and x identified(-1 each error or omission)	B2	
		(ii) (-)ve because a and x in opposite directions OR a directed towards mean position/centre.....	B1	[3]
	(b)	(i) forces in springs are $k(e + x)$ and $k(e - x)$ resultant = $k(e + x) - k(e - x)$ = $2kx$	C1 M1 A0	[2]
		(ii) $F = ma$ $a = -2kx/m$ (-)ve sign explained.....	B1 A0 B1	[2]
		(iii) $\omega^2 = 2k/m$ $(2\pi f)^2 = (2 \times 120)/0.90$ $f = 2.6 \text{ Hz}$	C1 C1 A1	[3]
	(c)	atom held in position by attractive forces atom oscillates, not just two forces OR 3D not 1D force not proportional to x <i>any two relevant points, 1 each, max 2</i>	B2	[2]
3	(a)	$pV/T = \text{constant}$	C1	